

ST 3011
Group Project



YOUTH UNEMPLOYMENT PATTERNS IN SRI LANKA

Group 6

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Introduction

Youth unemployment is a major socio-economic issue in Sri Lanka, especially in a period of economic uncertainty and changes in the labour market. High levels of unemployment among young people can lead to long-term income problems, loss of skills, and social challenges. Therefore, understanding the patterns and factors related to youth unemployment is important for informed decision-making.

This study examines youth unemployment patterns in Sri Lanka using data from the Sri Lanka Labour Force Survey (LFS) 2023. The analysis focuses on how unemployment varies across key demographic, educational, and geographic characteristics of the youth population.

The main aim of this study is to provide a clear understanding of youth unemployment in Sri Lanka and to identify the key factors associated with unemployment among youth.

Dataset Description.

This study uses data from the **Sri Lanka Labour Force Survey (LFS) 2023**, conducted by the Department of Census and Statistics (DCS). The Labour Force Survey is a nationally representative household survey designed to collect information on labour market participation, employment, and unemployment in Sri Lanka.

The dataset contains 18,937 observations and includes a wide range of socio-economic variables. These variables cover demographic characteristics such as age, sex, district, and residential sector, as well as education-related and employment-related information including educational attainment, employment status, occupation, industry, income, and working hours.

Both categorical variables (such as employment status, education level, sex, and sector) and numerical variables (such as age and income) are included in the dataset. This makes the dataset suitable for analysing youth unemployment patterns and examining variations across demographic, educational, and geographic characteristics.

Problem Definition

Youth unemployment in Sri Lanka varies across demographic, educational, and geographic groups. However, these variations are not clearly understood using overall unemployment figures alone. This study aims to analyse youth unemployment and identify the key factors associated with unemployment among youth using data from the Labour Force Survey 2023.

Definitions:

Youth :

This study defines youth as individuals aged 15–29 years in order to capture a broader group of young people, particularly those completing higher education and entering the labour market.

Unemployed :

Persons available and /or looking for work , and who did not work but taken steps during last 4 weeks to find a job and ready to accept a job if given a work opportunity within next two weeks.

Labour Force:

Labour Force is the population that is economically active, consisting of employed persons and unemployed persons who are actively seeking work.

Objectives

1. To describe youth unemployment patterns in Sri Lanka using demographic characteristics such as age, sex, sector and district.
2. To analyze the relationship between educational attainment and gender differences among unemployed youth.
3. To examine gender differences in youth unemployment across sectors and geographic areas in Sri Lanka.
4. To develop and evaluate a statistical model to examine factors associated with youth unemployment in Sri Lanka.

Data Preprocessing

The raw dataset obtained from the Sri Lanka Labour Force Survey (LFS) 2023 initially contained 18,937 observations with 128 variables. As a first step, only variables relevant to the objectives of the study were selected, including demographic characteristics, education-related variables, and employment-related questionnaire responses. This reduced the dataset to a manageable subset of key variables required for the analysis.

Several questionnaire-based variables were originally recorded as coded values. These variables were converted to numeric format where necessary and subsequently recoded into meaningful categorical labels to improve interpretability. In particular, variables such as sex, residential sector, district, education level, and current education or training status were recoded using official survey definitions.

To focus on the target population, the dataset was filtered to include only individuals aged **15–29 years**, in accordance with the study's definition of youth. This resulted in a youth sample of 3,949 individuals.

Employment status was derived using responses from multiple labour force survey questions. Individuals were classified as employed if they reported working during the reference period, unemployed if they were not working but actively seeking employment and available to work, and not in the labour force otherwise. This classification produced three distinct labour force categories for the youth population.

For subsequent analysis, individuals classified as not in the labour force were excluded, resulting in a final youth labour force dataset consisting of 1,428 individuals, of whom 1,239 were employed and 189 were unemployed. Additional checks confirmed that the dataset contained no duplicate records, and missing values were assessed to ensure data consistency. These preprocessing steps ensured that the final dataset was clean, well-defined, and suitable for descriptive analysis and statistical modelling of youth unemployment in Sri Lanka.

Final Variables Used After Preprocessing:

1. MONTH	13. CUEDU
2. SECTOR	14. P21
3. DISTRICT	15. Q2
4. PSU	16. Q4
5. HUNIT	17. Q6_A
6. HHOLD	18. Q47
7. SERNO	19. Q48
8. SEX	20. Q50
9. BYEAR	21. Q51
10. BMONTH	22. Q52
11. AGE	23. Employment Status
12. EDU	

The variables listed above include both analysis variables and survey identification variables. Variables such as PSU, HUNIT, HHOLD, and SERNO were used only for survey structure and data management purposes and were not directly used in the analysis. Questionnaire variables (e.g., Q2, Q4, Q6, Q47–Q52) were used to derive key indicators rather than being analysed individually.

Employment status was constructed from relevant questionnaire responses and used as the response variable in this study. Demographic and education-related variables including age, sex, residential sector, geographic location, educational attainment, and current education or training status, were used as explanatory variables to analyse youth unemployment patterns and to develop the statistical model.

Descriptive Analysis of Youth Unemployment

Here we will explore the distributions of variables and describe youth unemployment patterns in Sri Lanka using demographic characteristics such as age, sex, sector and district.

Target Variable: Employment Status

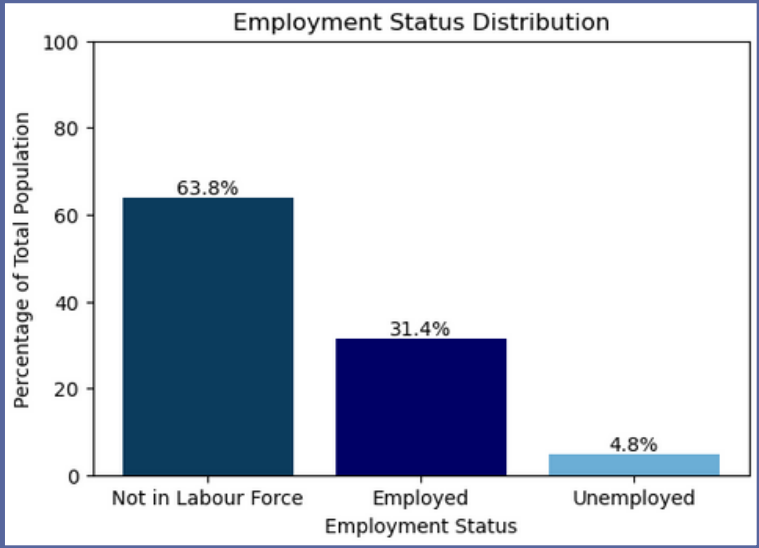


Figure 1: Employment status.

The figure illustrates the distribution of the population by employment status. A majority of individuals (63.8%) are classified as not in the labour force, meaning that many individuals are not currently engaged in work or job searching. Around 31.4% are employed, while only a small proportion (4.8%) are unemployed. Overall, the results indicate low labour force participation, with unemployment accounting for a relatively small share of the total population.

Although a large proportion of youth are classified as not in the labour force, the subsequent analysis focuses only on youth who are economically active (employed or unemployed). Individuals not in the labour force are excluded from further analysis, as youth unemployment is defined within the labour force population.

AGE

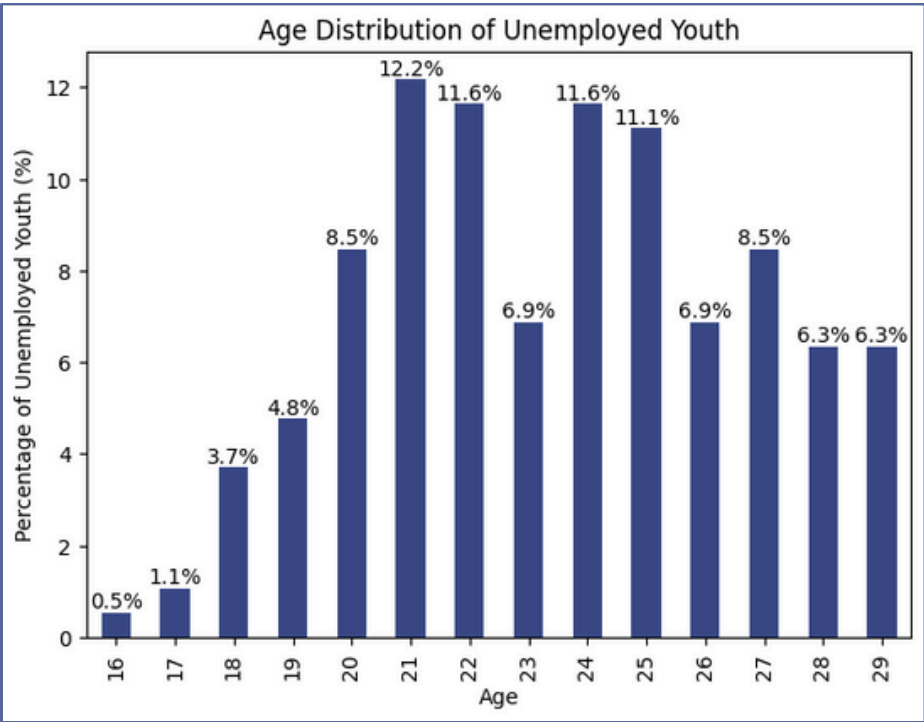


Figure 2: Age Distribution of Unemployed Youth

The graph shows that youth unemployment in Sri Lanka is highest among the youngest age group, especially those aged 19–24. This is mainly because many young people enter the labour market during this period after completing school or higher education and often struggle to find jobs that match their skills. As age increases toward 29, unemployment steadily declines, suggesting that with time, experience, and skill development, young people are more likely to secure employment or adjust their expectations to available opportunities.

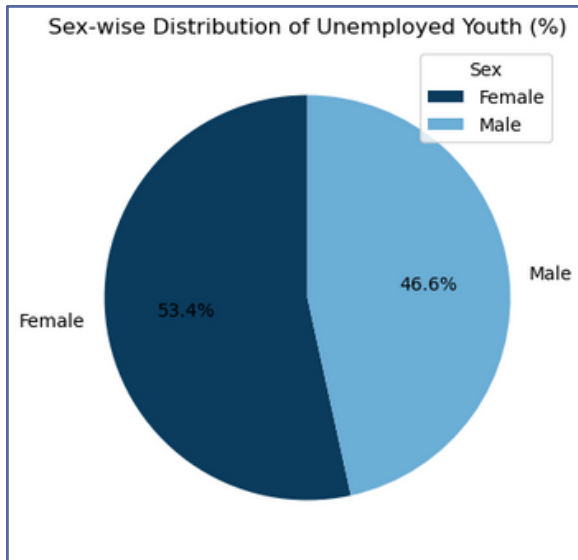


Figure 3: Sex-wise Distribution of Unemployed Youth.

SEX

The pie chart shows that youth unemployment is slightly higher among females (53.4%) compared to males (46.6%). This suggests that young women may experience more difficulty finding jobs than young men, possibly due to social expectations, family responsibilities, restricted mobility, or limited access to suitable employment opportunities.

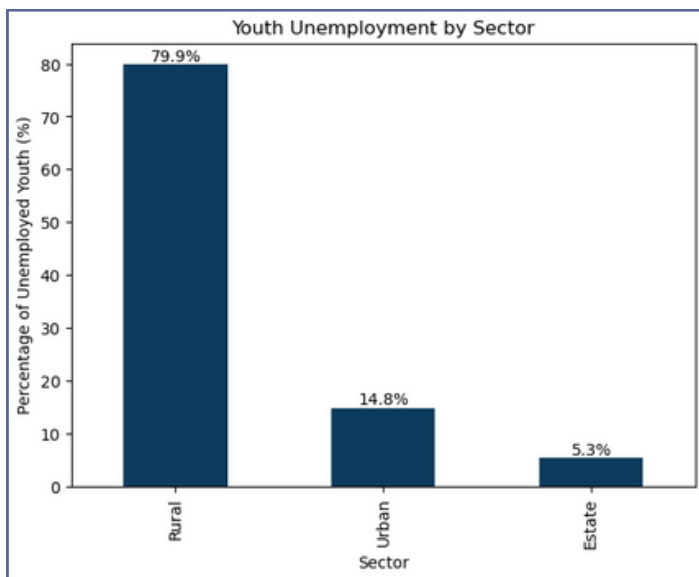


Figure 4: Sector-wise Distribution of Unemployed Youth.

SECTOR

The sector-wise bar chart shows that the majority of unemployed youth are from the rural sector (79.9%), while a much smaller proportion is from urban areas (14.8%) and estate sectors (5.3%). This shows a clear gap between rural and urban areas, where young people in rural communities struggle more to find jobs because there are fewer industries and limited employment opportunities available to them.

DISTRICT

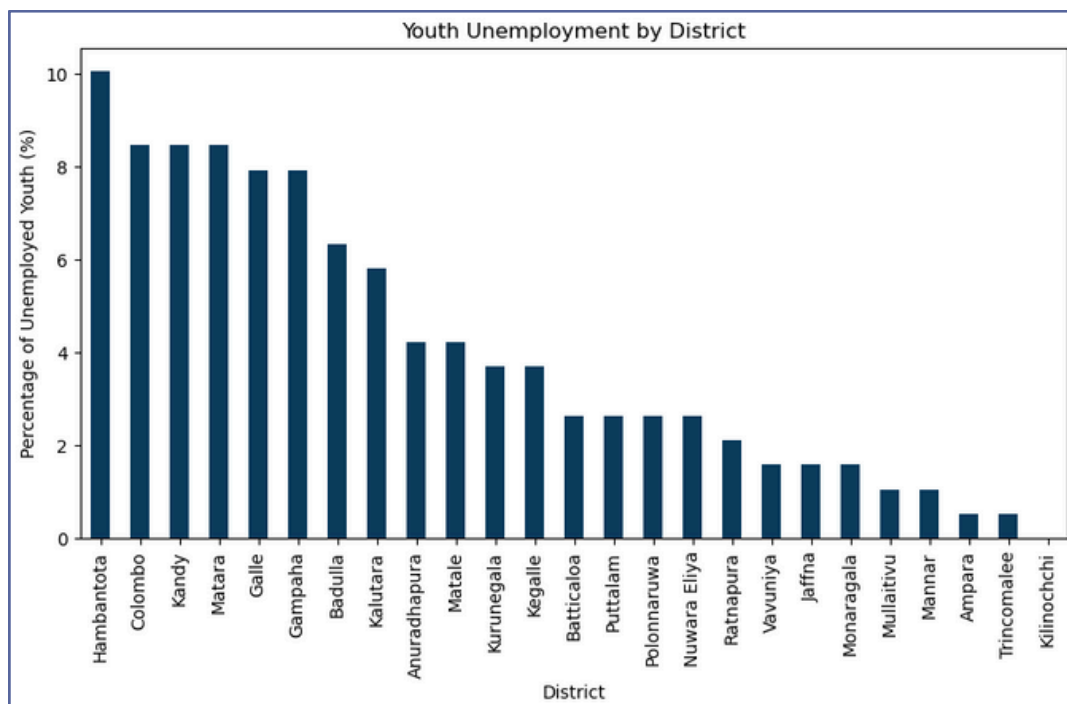


Figure 5: District-wise Distribution of Unemployed Youth.

The district-level analysis shows notable variation in youth unemployment across Sri Lanka. Hambantota records the highest percentage, followed by Colombo, Kandy, and Matara, while districts such as Kilinochchi, Trincomalee, and Ampara report the lowest levels. These differences reflect regional inequalities in economic activity, access to education, and availability of employment opportunities.

Educational Attainment and Youth Unemployment

Educational Levels (EDU)

There are 18 categories in EDU (educational attainment levels) variable. Therefore, To facilitate clearer analysis and interpretation, these categories were recoded into five broader educational groups based on the stage of formal education completed, as shown below.

Original EDU (Educational) Levels	Grouped Category
No schooling, Studying/ Studied Grade 1, Passed Grade 2, Passed Grade 3, Passed Grade 4, Passed Grade 5	Grade 5 or Below
Passed Grade 6, Passed Grade 7, Passed Grade 8, Passed Grade 9, Passed Grade 10	Grade 6-10
Passed G.C.E.(O/L)/ N.C.G.E.	G.C.E.(O/L)
Passed Grade 12, Passed G.C.E.(A/L)/ H.N.C.E.	G.C.E.(A/L)
Degree, Passed G.A.Q./G.S.Q., Post Graduate Degree/ Diploma, Special educational Institutions	Degree / Education institue

The distribution of unemployed youth across different educational levels is shown in the figure below.

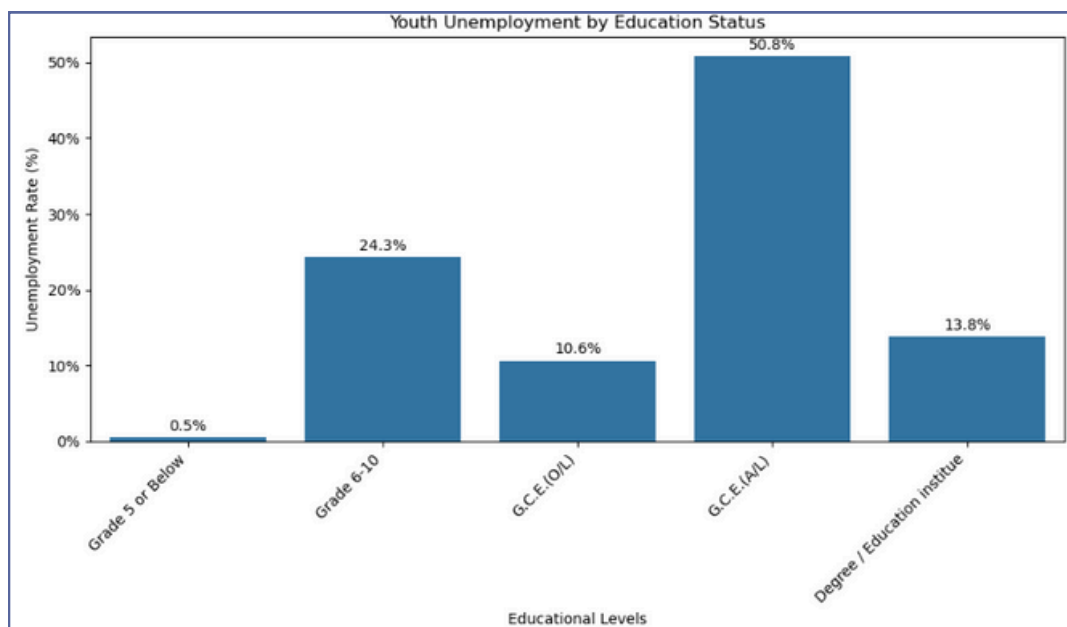


Figure 6: Youth Unemployment by Education Status..

The highest proportion of unemployed youth is observed among those who have completed G.C.E. (A/L), accounting for approximately 50.8% of the total unemployed youth population. This indicates that unemployment is most prevalent among youth who have completed A/L.

Current Educational attainment (CUEDU)

The following bar chart illustrates the unemployment rate among youth based on their current education or training status.

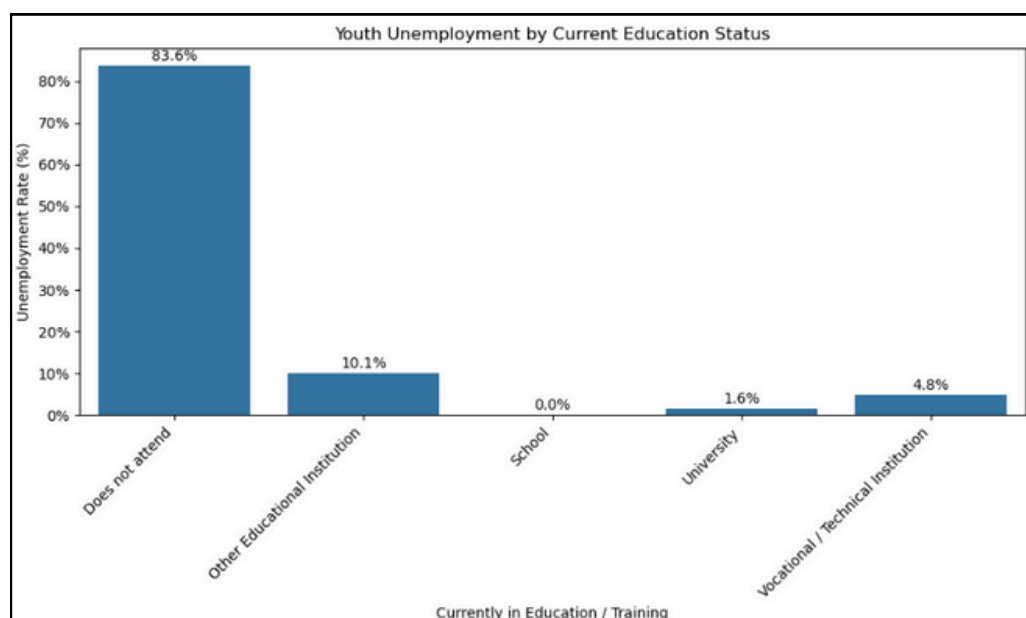


Figure 7: Youth Unemployment by Current Education Status..

The majority of unemployed youth do not attend any education or training institution (83.6%). Smaller proportions are observed among those attending other educational institutions (10.1%) and vocational/technical institutions (4.8%), while unemployment among university students is minimal (1.6%) and no unemployed youth are recorded among those currently attending School (0.0%).

Gender Differences by Educational Levels

Here we will examine youth unemployment in Sri Lanka in 2023, focusing on differences between males and females, across Educational levels.

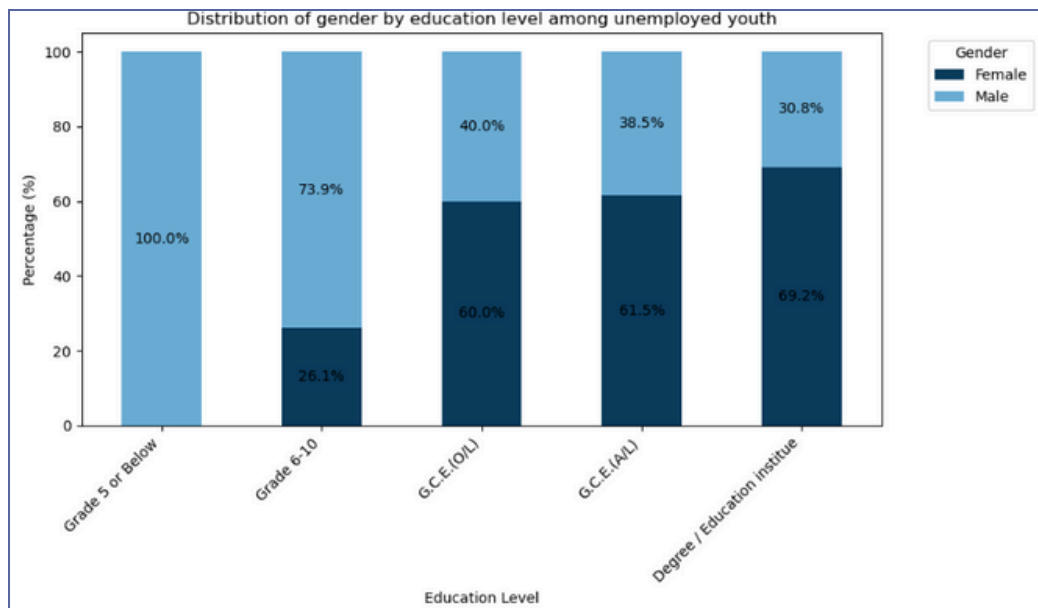


Figure 8: Distribution of gender by education level among unemployed youth.

At the Grade 5 or Below level, unemployment is observed only among male youth (100%), with no unemployed females recorded in this category. At the Grade 6–10 level, male youth account for a larger proportion of the unemployed population (73.9%) compared to females (26.1%). For G.C.E. (O/L) , G.C.E. (A/L) and Degree / Education Institute education levels, female youth constitute the majority of the unemployed population. Overall, female unemployment is more prevalent across most educational levels, particularly at secondary and higher education levels.

Statistical analysis

Chi-square tests of independence were conducted to determine whether Educational Levels associated with gender among unemployed youth.

H_0 : There is no association between EDU (Educational Levels) and gender

H_1 : There is an association between EDU (Educational Levels) and gender

Degree of freedom: 3
Chi-square: 20.14426450994629
p-value: 0.00015844856721700787

Since p-value is less than 0.05 significance level, we reject H_0 . Therefore, at 5% significance level, there is sufficient evidence to conclude that a statistically significant relationship exists between education level (EDU) and gender.

Gender Differences by Current Education

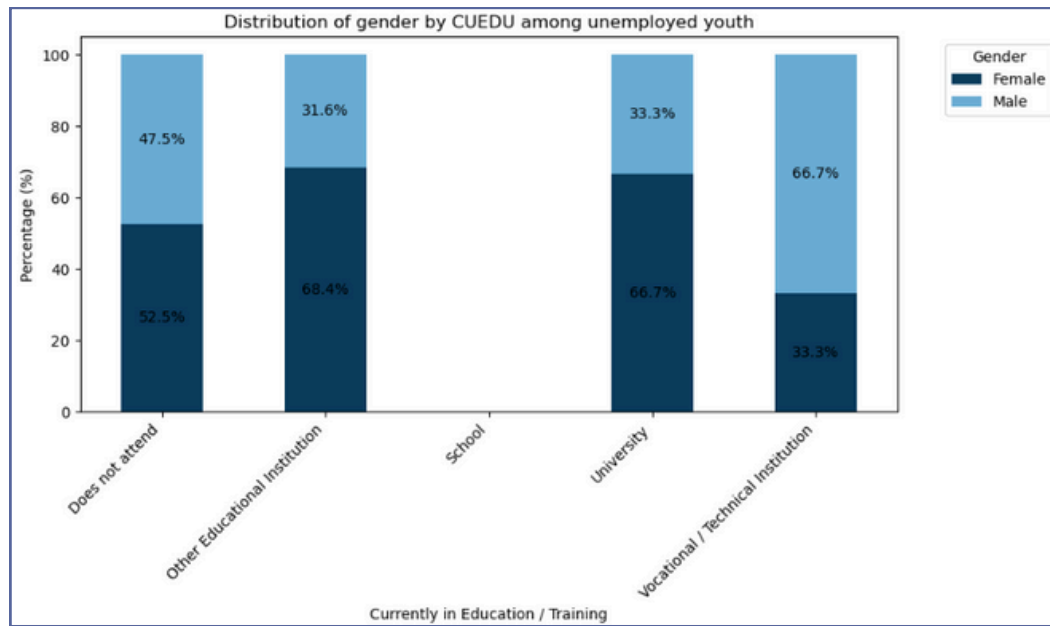


Figure 9: Distribution of gender by CUEDU among unemployed youth..

Above figure illustrates the distribution of gender by current education or training status among unemployed youth. Among those who do not attend any institution, females account for a slightly higher proportion (52.5%) than males (47.5%). Female unemployment is notably higher among youth attending other educational institutions (68.4%) and universities (66.7%). In contrast, males constitute the majority of unemployed youth in vocational/technical institutions (66.7%).

Statistical analysis

Chi-square tests of independence were conducted to determine whether Educational Levels associated with gender among unemployed youth.

H_0 : There is no association between CUEDU (Current Educational attainment) and gender

H_1 : There is an association between CUEDU (Current Educational attainment) and gender

Degree of freedom: 2
Chi-square: 2.434682852445804
p-value: 0.2960161023747797

Since the p-value is greater than the 0.05 significance level, we do not reject H_0 . Therefore, there is no statistically significant association between current educational status and gender among unemployed youth at 5% significance level.

Gender Differences in Youth Unemployment Across Sectors and Districts

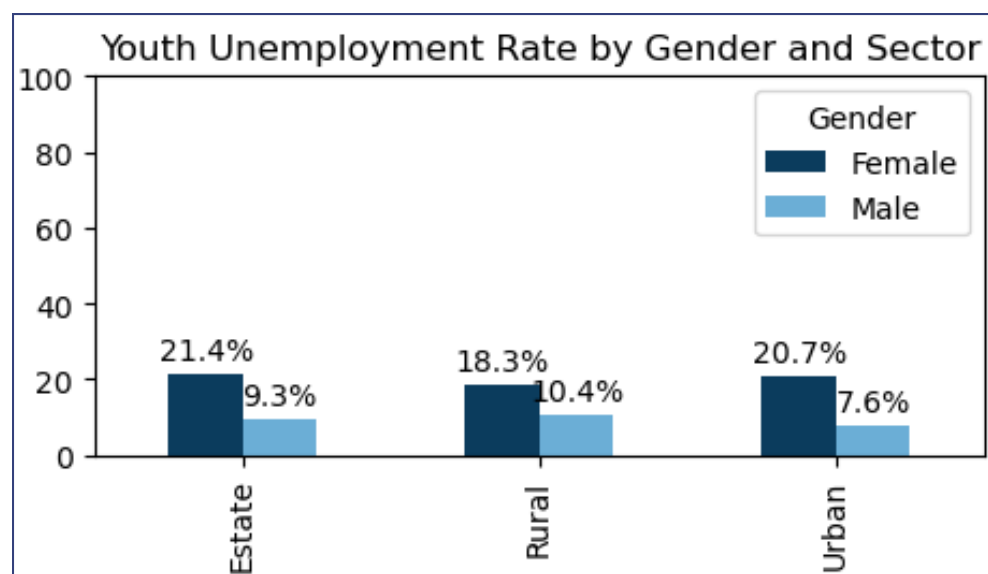
Here we will examine youth unemployment in Sri Lanka, focusing on differences between males and females, across sectors (urban, rural, estate) and districts.

Youth Unemployment by Gender and Sector

Here illustrates youth unemployment rates by gender across the three sectors.

	employment_s	Employed	Unemployed
SECTOR	SEX		
Estate	Female	78.57	21.43
	Male	90.7	9.3
Rural	Female	81.69	18.31
	Male	89.63	10.37
Urban	Female	79.27	20.73
	Male	92.41	7.59

Percentages are calculated within each gender-sector group and only unemployed youth are represented.



Female youth experience higher unemployment than male youth across all sectors.

Figure 10: Youth unemployment (%) by gender across sectors (Urban, Rural, Estate).

Statistical Analysis

Chi-square tests of independence were conducted to determine whether unemployment status is associated with gender in each sector. Results are summarized below:

H_0 : Gender is independent of unemployment status among youth in the sector.

H_1 : Gender is associated with unemployment status among youth in the sector.

Sector	Chi-square	df	p-value	Significant or Not
Urban	7.2	1	0.0073	Significant
Rural	13.78	1	0.0002	Significant
Estate	1.18	1	0.2773	Not Significant

Although the graph shows higher female youth unemployment in the Estate sector, the Chi-square test indicates that this difference is not statistically significant. This is likely due to the small number of unemployed youth and low expected cell frequencies in the Estate sector, which reduces the power of the test. Therefore, Estate sector results should be interpreted with caution.

Interpretation:

- Gender differences in youth unemployment are statistically significant in urban and rural sectors, with female youth experiencing higher unemployment than males.
- The estate sector does not show a significant gender gap, suggesting relatively balanced unemployment rates between males and females in this sector.

Youth Unemployment by Gender and District

	employment_status	Employed	Unemployed	District_Unemployment_Rank
DISTRICT	SEX			
Matara	Female	57.14	42.86	1
	Male	72.97	27.03	1
Badulla	Female	69.23	30.77	2
	Male	65.22	34.78	2
.	.	.	.	.
.	.	.	.	.

District-level analysis identifies top 10 districts with the highest total youth unemployment (male + female). Graph shows unemployment rates by gender for these districts.

- Districts such as Matara, Badulla and Hambantota exhibit the highest youth unemployment rates.
- In most districts, female youth unemployment is higher than male youth unemployment, reflecting persistent gender disparities across regions.
- This geographic variation highlights the influence of local labour market conditions on youth employment.

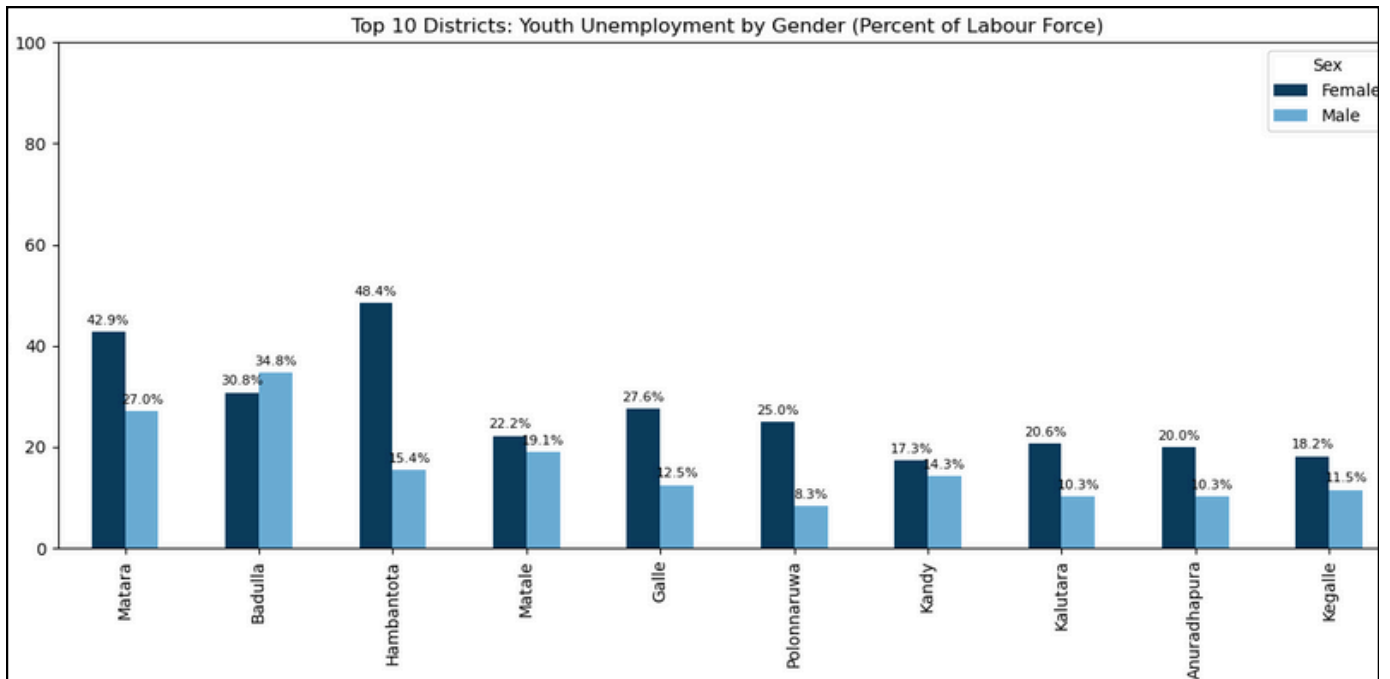


Figure 11: Youth unemployment (%) by gender for the top 10 districts, ordered by total unemployment

Statistical Analysis

H_0 : Gender is independent of unemployment status among youth in the district.

H_1 : Gender is associated with unemployment status among youth in the district.

Chi-square tests were also conducted for the top 10 districts by total youth unemployment. Significant gender differences were observed in districts such as:

- Hambantota

Other districts did not show statistically significant gender differences, indicating that gender disparities are district-specific.

District: Hambantota
 Chi-square statistic = 5.52
 Degrees of freedom = 1
 p-value = 0.0187

Interpretation:

- Gender disparities exist but depend on district.

```
*** Chi-square Assumption Check: District Level
-----
Colombo: ✓ min expected = 6.16
Gampaha: ✓ min expected = 5.79
Kalutara: ✓ min expected = 5.12
Kandy: ✓ min expected = 7.76
Matale: ⚠ min expected = 3.69 (assumption violated)
Nuwara Eliya: ⚠ min expected = 2.16 (assumption violated)
Galle: ✓ min expected = 5.12
Matara: ⚠ min expected = 4.39 (assumption violated)
Hambantota: ✓ min expected = 8.67
Jaffna: ⚠ min expected = 0.70 (assumption violated)
Mannar: ⚠ min expected = 0.38 (assumption violated)
Vavuniya: ⚠ min expected = 1.05 (assumption violated)
Mullaitivu: ⚠ min expected = 0.53 (assumption violated)
Kilinochchi: skipped (incomplete table)
Batticaloa: ⚠ min expected = 1.34 (assumption violated)
Ampara: ⚠ min expected = 0.38 (assumption violated)
Trincomalee: ⚠ min expected = 0.27 (assumption violated)
Kurunegala: ⚠ min expected = 2.84 (assumption violated)
Puttalam: ⚠ min expected = 1.25 (assumption violated)
```

Chi-square tests at the district level revealed that expected cell frequencies were adequate only for a limited number of districts, mainly those with larger sample sizes. For several districts, expected counts were below the recommended threshold of five, particularly in regions with smaller youth labour force samples. Therefore, district-level inferential results are interpreted with caution, and emphasis is placed on descriptive patterns and sector-level statistical tests.

Youth Unemployment by Gender and Province

District-level Chi-square tests revealed violations of expected frequency assumptions for several districts due to small sample sizes. Therefore, inferential analysis was conducted at the province level by aggregating districts into provinces, ensuring adequate expected cell counts and more reliable statistical inference.

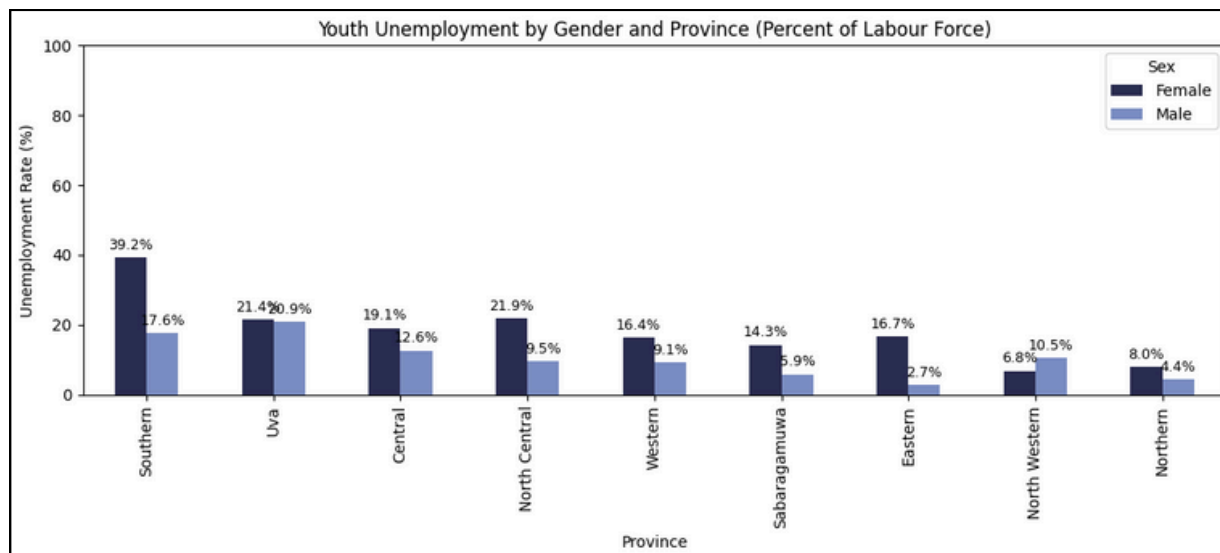


Figure 12: Youth unemployment (%) by gender and province

Female youth unemployment is higher than male unemployment in most provinces of Sri Lanka.

Uva and Southern provinces show the highest youth unemployment rates.

The size of the gender gap in unemployment differs across provinces.

Statistical Analysis

H_0 : Gender is independent of unemployment status among youth in the province.

H_1 : Gender is associated with unemployment status among youth in the province.

```
Province: Southern
Chi-square statistic = 9.94
Degrees of freedom = 1
p-value = 0.0016
```

```
Province: Eastern
Chi-square statistic = 4.5
Degrees of freedom = 1
p-value = 0.0339
```

Chi-square tests indicate that female youth unemployment is significantly higher than male youth unemployment in Southern and Eastern provinces. In other provinces, the gender difference is not statistically significant, suggesting that gender disparities vary regionally.

Modelling Factors Associated with Youth Unemployment

Objective Overview

The objective of this analysis is to identify key socio-demographic, educational, and regional factors associated with youth unemployment in Sri Lanka. A statistical modelling approach was adopted to quantify the relationship between these factors and the likelihood of being unemployed among youth.

Data and Response Variable

The analysis was conducted using the youth labour force dataset, which includes individuals aged 15–29 years who are either employed or unemployed. Youth who were not in the labour force were excluded from this objective, as unemployment is defined only among labour force participants.

The response variable was defined as a binary indicator:

0 – Employed

1 – Unemployed

Explanatory Variables

Based on relevance to youth labour market outcomes and interpretability, the following explanatory variables were selected:

Age

Sex

Residential sector (Urban, Rural, Estate)

Current Education/Trainin

Grouped educational attainment

Province – derived by aggregating districts

Educational attainment was grouped into major levels to reduce sparsity and improve model stability and interpretation. Provinces were used instead of districts to capture regional variation while avoiding excessive categorical complexity.

All categorical variables were converted into dummy variables, and reference categories were automatically set using `drop_first = True`.

Model Specification

A binary logistic regression model was used to estimate the probability of youth unemployment. Logistic regression is appropriate due to the binary nature of the response variable and its interpretability in socio-economic studies.

Two models were estimated:

1. Baseline logistic regression model

2. Weighted logistic regression model, where class weights were applied to address imbalance between employed and unemployed youth.

The dataset was split into:

70% training data

30% testing data

Model Evaluation and Comparison

Model performance was evaluated using:

Classification accuracy

Classification report (precision, recall, F1-score)

Confusion matrices

Model Accuracy:

```
Baseline model accuracy: 0.8555
Weighted model accuracy: 0.6270
```

The baseline logistic regression model achieved an accuracy of 85.55%, while the weighted logistic regression model achieved an accuracy of 62.70%. Although the baseline model shows higher accuracy, this is mainly due to the large proportion of employed individuals in the dataset and does not indicate good performance in identifying unemployed youth.

Baseline Logistic Regression – Classification Report:

... Baseline Logistic Regression					
	precision	recall	f1-score	support	
0	0.86	0.99	0.92	370	
1	0.00	0.00	0.00	59	
accuracy			0.86	429	
macro avg	0.43	0.50	0.46	429	
weighted avg	0.74	0.86	0.80	429	

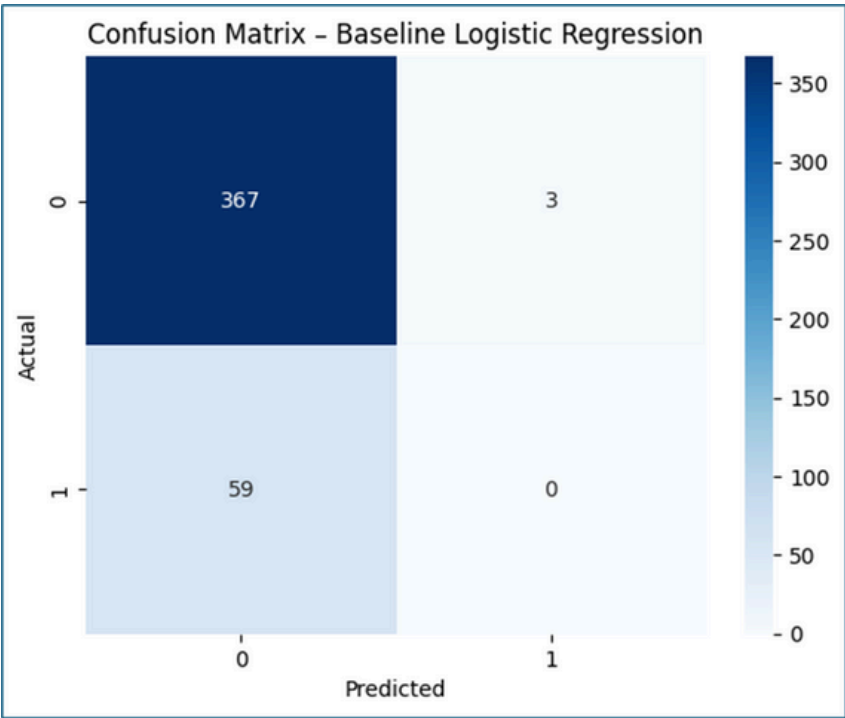
For the baseline model, the precision and recall for employed youth (class 0) are 0.86 and 0.99, respectively. However, the model completely fails to identify unemployed youth (class 1), with precision, recall, and F1-score equal to 0.00, indicating poor performance for the target group.

Weighted Logistic Regression – Classification Report:

Weighted Logistic Regression					
	precision	recall	f1-score	support	
0	0.93	0.62	0.74	370	
1	0.22	0.69	0.34	59	
accuracy			0.63	429	
macro avg	0.58	0.66	0.54	429	
weighted avg	0.83	0.63	0.69	429	

The weighted logistic regression model improves the identification of unemployed youth, achieving a recall of 0.69 for the unemployed class. Although overall accuracy decreases to 62.70%, the model provides a more balanced classification and better captures unemployment cases.

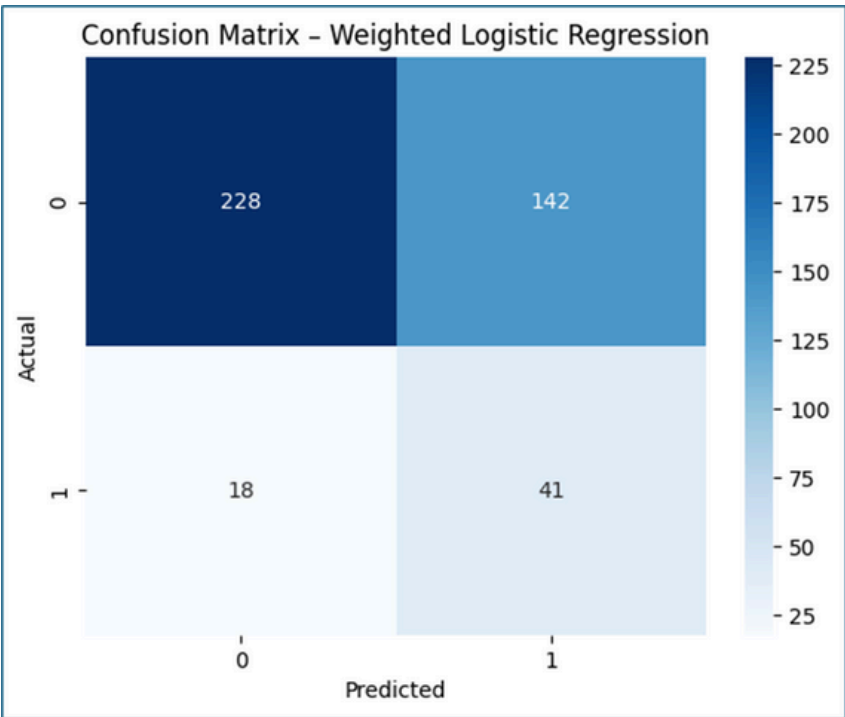
Confusion Matrix – Baseline Logistic Regression:



Out of 59 unemployed youth, the baseline model fails to correctly classify any as unemployed. Almost all observations are predicted as employed (367 correctly classified employed individuals), confirming the model's bias toward the majority class.

Figure 13: Confusion Matrix - Baseline Logistic Regression .

Confusion Matrix – Weighted Logistic Regression



The weighted model correctly identifies 41 out of 59 unemployed youth, while 18 are misclassified. Although some employed individuals are incorrectly predicted as unemployed, the model performs significantly better in detecting unemployment.

Figure 14: Confusion Matrix - Weighted Logistic Regression .

Final Model Selection

Due to the imbalance between employed and unemployed youth in the sample, the baseline logistic regression model showed poor performance in identifying unemployed individuals. Therefore, a weighted logistic regression model was selected, as it improves the identification of unemployed youth and provides more meaningful interpretation despite a lower overall accuracy.

Interpretation of Logistic Regression Coefficients

Variable	Coefficient
P21_Not studying	-0.499388
PROVINCE_Southern	0.898835
PROVINCE_Uva	0.540064
PROVINCE_North Central	-0.021922
PROVINCE_Western	-0.153975
PROVINCE_Sabaragamuwa	-0.545471
PROVINCE_Eastern	-0.628769
PROVINCE_North Western	-0.638001
PROVINCE_Northern	-1.396793
SECTOR_Rural	-0.228195
SECTOR_Urban	-0.44768
EDU_grouped_G.C.E.(A/L)	-0.429
EDU_grouped_Grade 5 & Below	-0.533145
EDU_grouped_G.C.E.(O/L)	-0.914667
EDU_grouped_Grade 6–10	-0.95446
SEX_Male	-0.506652
AGE	-0.102653

The coefficients in table are interpreted relative to the reference categories: female for sex, currently studying/training for education status (P21), degree/education institute for educational attainment, Central Province for province, and estate sector for residential sector.

Positive coefficient values indicate a higher likelihood of youth unemployment compared to the reference group, while negative values indicate a lower likelihood. Youth residing in the Southern Province ($\beta = 0.899$) and Uva Province ($\beta = 0.540$) show a higher probability of being unemployed compared to youth in the Central Province. In contrast, youth in the Western Province ($\beta = -0.154$) have a lower likelihood of unemployment, reflecting better labour market opportunities in more urbanised regions.

The negative coefficient for SEX_Male ($\beta = -0.507$) indicates that female youth are more likely to be unemployed than male youth. Sectoral effects show that youth living in urban ($\beta = -0.448$) and rural ($\beta = -0.228$) sectors have a lower unemployment risk compared to those in the estate sector.

Educational attainment coefficients suggest that youth with lower educational qualifications face higher unemployment risk compared to those with degree-level education. The negative coefficient for Age ($\beta = -0.103$) indicates that the likelihood of unemployment decreases as age increases within the youth population.

Conclusions

- Youth unemployment in Sri Lanka remains a significant issue, particularly among individuals aged 15–29 years, with the highest unemployment observed in the 19–24 age group.
- Female youth experience higher unemployment rates than male youth across most sectors and regions, indicating persistent gender disparities in the labour market.
- A large proportion of unemployed youth are concentrated in rural areas, highlighting limited employment opportunities outside urban centres.
- Youth unemployment shows considerable regional variation, with higher levels observed in districts such as Hambantota and in provinces including Southern and Uva.
- Educational attainment is strongly associated with youth unemployment, with the highest proportion of unemployed youth having completed G.C.E. (A/L), suggesting a mismatch between education outcomes and labour market requirements.
- Chi-square analysis reveals a statistically significant association between completed education level and gender among unemployed youth, while no significant association is found between current education or training status and gender.
- Logistic regression results indicate that younger age, female gender, residence in the estate sector, lower educational attainment, and residence in Southern and Uva provinces increase the likelihood of youth unemployment.

Overall, youth unemployment in Sri Lanka is a multidimensional issue influenced by demographic, educational, and geographic factors, highlighting the need for targeted policies to improve education-to-employment transitions, promote gender equity, and expand employment opportunities in disadvantaged regions.

Limitations

- The study relies on secondary data from the Sri Lanka Labour Force Survey (LFS) 2023, limiting the analysis to available variables and preventing inclusion of factors such as work experience and job preferences.
- The analysis considers only economically active youth, excluding those not in the labour force, which may limit the generalizability of the findings.
- Some district-level analyses were constrained by small sample sizes, leading to low expected frequencies in chi-square tests and requiring cautious interpretation.
- Educational categories were grouped for analytical simplicity, which may mask variations within detailed education levels.
- Class imbalance between employed and unemployed youth affected model performance; although weighting improved results, some misclassification remains.
- As the data are cross-sectional, causal relationships and time trends in youth unemployment cannot be established.

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<https://drive.google.com/file/d/1AqADUAmnTsAv8b6EHTaGcWy8PYfmsfls/view?usp=sharing>